

## Mark schemes

### Q1.

(a)  $y(0) = 0;$

$$y(1) = \frac{1}{3} = 0.3$$

$$y(2) = \frac{1}{3} = 0.3$$

$$y(3) = \frac{3}{11} = 0.27$$

$$y(4) = \frac{4}{18} = 0.2$$

*all 5 x-values PI by 5 correct y-values*

B1

*at least 4 y-values exact or rounded or truncated to at least 4sf*

B1

$$\frac{1}{3} \times 1(0 + 0.2 + 4[0.3 + 0.27] + 2[0.3])$$

*correct use of Simpson's rule using  $\frac{1}{3}$  and 4 and 2 correctly with candidate's 5 y-values*

M1

$$= 1.104$$

*CAO (must be exactly this value)*

A1

4

(b)  $\int_0^4 \frac{x}{x^2 + 2} dx = \frac{1}{2} [\ln(x^2 + 2)]$

*for  $k \ln(x^2 + 2)$*

M1

*all correct; limits not needed*

A1

$$= \frac{1}{2} (\ln 18 - \ln 2)$$

*For  $k (\ln 18 - \ln 2)$*

A1F

$$= \frac{1}{2\ln 9}$$

combining candidate's logarithms correctly (must be seen)

A1F

$$= \ln 3$$

CAO (must be exactly this) NMS scores 0 / 5

A1

5

[9]

**Q2.**

(a)

|       |                    |                    |                   |                     |                     |
|-------|--------------------|--------------------|-------------------|---------------------|---------------------|
| $x_i$ | $0.4(\frac{2}{5})$ | $1.2(\frac{6}{5})$ | $2(\frac{10}{5})$ | $2.8(\frac{14}{5})$ | $3.6(\frac{18}{5})$ |
| $y_i$ | 5.20231            | 5.35985            | 5.91608           | 6.99657             | 8.58231             |

All 5 x-values correct, PI by 5 correct y-values.

B1

At least 4 correct y-values rounded or truncated to at least 4 s.f. or in surd form  $\sqrt{27 + (0.4)^3}$ ,  $\sqrt{27 + (1.2)^3}$ , etc. or  $\sqrt{27.064}$ ,  $\sqrt{28.728}$ , etc. or sight of 32.057...

B1

$$\int_0^4 \sqrt{27 + x^3} \approx 0.8 \sum_{i=1}^5 y_i$$

Correct use of mid-ordinate rule using 0.8 with candidate's 5 y-values.

Dependent on first B1

M1

$$(= 0.8 \times 32.057...)$$

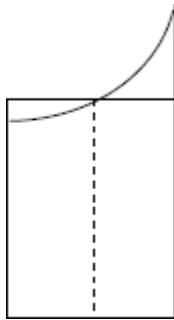
$$= 25.6$$

CAO (must be exactly this) and no error seen

A1

4

(b)



Could be gained without answering part (a)  
Diagram showing curve through the midpoint of the top of rectangle. May have one or more rectangles.

B1

“Smaller” OE

Dependent on B1

E1

2

[6]

Q3.

| $x$            | $y$ |
|----------------|-----|
| 0              | 1   |
| $\frac{1}{2}$  | 2   |
| 1              | 4   |
| $1\frac{1}{2}$ | 8   |
| 2              | 16  |
| $2\frac{1}{2}$ | 32  |
| 3              | 64  |

all 7x values correct (and no extra)  
(PI by 7 correct y values)

B1

5 or more correct y values, exact  $\left(4^{\frac{1}{2}}, 4^1, \dots\right)$  or evaluated  
(in table or in formula)

B1

$$A = \frac{1}{3} \times \frac{1}{2} [65 + 4 \times 42 + 2 \times 20]$$

correct substitution of their 7 y-values into Simpson's rule

M1

$$= \frac{91}{2} \text{ or } 45.5 \text{ or } \frac{273}{6}$$

CAO

A1

[4]

Q4.

| $x$ | $y$    |
|-----|--------|
| 0.5 | 3.9163 |
| 0.7 | 1.8748 |
| 0.9 | 0.9520 |
| 1.1 | 0.3773 |

All 4 correct  $x$  values (and no extras used)

B1

3 +  $y$  decimal values rounded or truncated to 2 dp or better  
(in table or in formula) (PI by correct answer)

M1

$$\int = 0.2 \times \sum y$$

Correct substitution of their 4  $y$  values (of which 3 are correct), either listed or totalled

m1

$$(\text{= } 0.2 \times 7.12 \dots) = 1.424$$

CAO

A1

[4]

Q5.

| $x$  | $y$                         |
|------|-----------------------------|
| 0.05 | $\cos \sqrt{1.15} = 0.4780$ |
| 0.15 | $\cos \sqrt{1.45} = 0.3585$ |
| 0.25 | $\cos \sqrt{1.75} = 0.2454$ |
| 0.35 | $\cos \sqrt{2.05} = 0.1386$ |

Using 4 correct  $x$ -values, PI

B1

At least 3 correct  $y$ -values,  
(condone unsimplified correct expressions),  
Or correct values rounded to 2 s.f. or  
truncated to 2 s.f.

M1

$$0.1 \times \Sigma y$$

*Used and must be working in radians*

m1

$$= 0.122$$

CAO

*Must be 3 s.f.*

A1

[4]

**Q6.**

(a)

| $x$  | $y$     |
|------|---------|
| 1.5  | 1.98100 |
| 4.5  | 3.22883 |
| 7.5  | 4.11496 |
| 10.5 | 4.74710 |

*x values correct PI*

B1

*3 + y values correct to 2sf or better or exact values*

M1

*1.981, 3.228 / 9, 4.114 / 5, 4.747 for y  
(or better)*

A1

$$\int = 3 \times \Sigma y = 42.2$$

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*(Note: 42.2 with evidence of mid-ordinate rule with four strips scores 4/4)*

A1

4

(b) (i)  $y = \ln(x^2 + 5)$

$$e^y = x^2 + 5$$

OE

$$x^2 = e^y - 5$$

*AG Must see middle line, and no errors*

B1

1

(ii)  $(\pi) \int (e^y - 5)(dy)$

*Condone omission of brackets around*

$f(y)$  throughout

M1

$$= (\pi) [e^y - 5y]_{(5)}^{(10)}$$

A1

$$= (\pi) [(e^{10} - 50) - (e^5 - 25)]$$

$$F(10) - F(5)$$

m1

$$V = \pi [e^{10} - e^5 - 25]$$

CSO

including correct notation – must see **dy**  
ISW if evaluated

A1

4

[9]

**Q7.**

(a)

| x    | y                          |
|------|----------------------------|
| 0.5  | $\frac{4}{9} = 0.\dot{4}$  |
| 0.75 | $\frac{48}{91} = 0.5275$   |
| 1    | $\frac{1}{2} = 0.5$        |
| 1.25 | $\frac{80}{189} = 0.4233$  |
| 1.5  | $\frac{12}{35} = 0.3429$   |
| 1.75 | $\frac{112}{407} = 0.2752$ |
| 2    | $\frac{2}{9} = 0.\dot{2}$  |

x values correct PI

B1

At least 5 y values that would be correct

to 2 sf or better, or exact values. May be seen within working.

B1

$$\left[ \left( \frac{4}{9} + \frac{2}{9} \right) + 4 \left( \frac{48}{91} + \frac{80}{189} + \frac{112}{407} \right) + 2 \left( \frac{1}{2} + \frac{12}{35} \right) \right]$$

$$\int = \frac{1}{3} \times 0.25 [ \quad ]$$

Clear attempt to use 'their' y values within Simpson's rule

M1

$$= 0.605$$

Answer must be 0.605 with no extra sf  
(Note 0.605 with no evidence of Simpson's rule scores 0/4)

A1

4

(b)

$$\int_0^1 \frac{x^2}{1+x^3} dx$$

$$= \frac{1}{3} \ln(1+x^3)$$

k ln (1 + x<sup>3</sup>) condone missing brackets

M1

Correct. A1 may be recovered for missing brackets if implied later

A1

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$$= \frac{1}{3} \ln(1+1) \left( -\frac{1}{3} \ln 1 \right)$$

F(1) (–F(0))

m1

$$= \frac{1}{3} \ln 2$$

ln 1 must not be left in final answer

A1

**Alternative**

$$u = 1 + x^3 \quad du = 3x^2 dx$$

$$\int = \int \frac{du}{3u}$$

$$\frac{dz}{dx} \text{ correct and integral of form } \int \frac{dz}{z}$$

(M1)

$$= \frac{1}{3} [\ln z]$$

(A1)

$$= \frac{1}{3} \ln 2 \left( -\frac{1}{3} \ln 1 \right)$$

Correct substitution of correct  $u$  limits or  
conversion back to  $x$  and  $F(1) (-F(0))$

(m1)

$$= \frac{1}{3} \ln 2$$

$\ln 1$  must not be left in final answer

(A1)

4

[8]

**Q8.**

| $x$ | $y$      |
|-----|----------|
| 1   | 0.5      |
| 3   | 0.366(0) |
| 5   | 0.309(0) |
| 7   | 0.247(3) |
| 9   | 0.25     |

$x$  values and no extra values

B1

$$4 + \text{correct } y \text{ values or } \frac{1}{1 + \sqrt{3}} \text{ etc}$$

B1

$$\int = \frac{1}{3} \times 2 \times \left[ \frac{(0.5 + 0.25) +}{4(0.3660 + 0.2743) + 2(0.3090)} \right]$$

Correct application of Simpson's rule  
for their  $x$  values ( $x$  odd)

M1

$$= 2.62$$

CSO must be 3sf



Q9.

(a)  $V = \pi \int x^2 dx$   
PI

B1

$$V = \frac{(\pi)}{4} \int (100 - y^2) dy$$

$k \int (100 - y^2) dy$  may be recovered

Allow  $\int (their\ x)^2 dx$ , expanded

M1

$$= \frac{(\pi)}{4} \left[ 100y - \frac{y^3}{3} \right]_{(0)}^{(10)}$$

A1

$$= \frac{(\pi)}{4} \left[ \frac{2000}{3} \right]$$

For  $F(10) - F(0)$

m1

$$= \frac{500\pi}{3}$$

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OE CSO

$$\left[ \begin{array}{l} \text{SC: if rotated about } x\text{-axis} \\ V = \pi \left[ 100x - \frac{4x^3}{3} \right]_0^5 \quad M1 \\ = \frac{1000}{3} \pi \quad A1 \text{ max } 2/5 \end{array} \right]$$

A1

(b)

| $x$ | $y$     |             |
|-----|---------|-------------|
| 0.5 | 9.95(0) | } or better |
| 1.5 | 9.539   |             |
| 2.5 | 8.66(0) |             |
| 3.5 | 7.141   |             |
| 4.5 | 4.359   |             |

Correct  $x$

B1

4 + correct  $y$  to 2sf

M1

All  $y$  correct

A1

$$A = 1 \times \sum y = 39.6$$

(39.6 scores  $\frac{4}{4}$ )

A1

4

[9]

Q10.

(a)

$$y = (e^{3x} + 1)^{\frac{1}{2}}$$

$$\frac{dy}{dx} = \frac{1}{2} (e^{3x} + 1)^{-\frac{1}{2}} \times 3e^{3x}$$

$$\frac{1}{2} (e^{3x} + 1)^{-\frac{1}{2}}$$

M1

$$e^{3x}$$

A1

$$\frac{3}{2} \text{ (allow } \frac{1}{2} \times 3 \text{) w.n.f.e}$$

A1

$x = \ln 2$ :

$$\frac{dy}{dx} = \frac{3}{2} (e^{\ln 8} + 1)^{-\frac{1}{2}} \times e^{\ln 8}$$

correct substitution into their  $\frac{dy}{dx}$

(must use  $\ln 8$  or  $\ln 2^3$ )

M1

$$= \frac{3}{2} \times \frac{1}{3} \times 8$$

$$= 4$$

A1

5

(b)

| $x$  | $y$      |
|------|----------|
| 0.25 | 1.765(5) |
| 0.75 | 3.238(5) |
| 1.25 | 6.597(1) |
| 1.75 | 13.84(1) |

correct  $x$  values

3 or 4 correct  $y$  values 4 s.f. or better

B1

B1

$$\int = 0.5 \times \sum y \quad \text{P.I.}$$

M1

$$= 12.7$$

sc 12.7 with no working  $\frac{2}{4}$

A1

4

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(c)

$$v = \pi \int y^2 dx$$

$$= (\pi) \int (e^{3x} + 1) (dx)$$

M1

$$= (\pi) \left[ \frac{1}{3} e^{3x} + x \right]_{(0)}^{(2)}$$

$$ke^{3x} + x$$

A1

$$= (\pi) \left[ \left( \frac{1}{3} e^6 + 2 \right) - \left( \frac{1}{3} e^0 + 0 \right) \right]$$

correct substitution into  $f \left( \int e^{3x} \right)$

m1

$$= \pi \left[ \frac{1}{3} e^6 + \frac{5}{3} \right]$$

$$\left( = \frac{\pi}{3} (e^6 + 5) \right)$$

CSO

A1

4

[13]

**Q11.**

$$x = 1.5, 2.5, 3.5, 4.5$$

Method

*x values*

$$\begin{array}{l} y_1 = 0.7115 \\ y_2 = 0.5218 \\ y_3 = 0.4439 \\ y_4 = 0.3993 \end{array} \quad \begin{array}{l} 0.712 \\ 0.522 \\ 0.444 \\ 0.399 \end{array} \quad \left. \vphantom{\begin{array}{l} y_1 \\ y_2 \\ y_3 \\ y_4 \end{array}} \right\} \text{AWRT}$$

3 correct y's

M1

A1

$$A = 1 \times (y_1 + y_2 + y_3 + y_4) = 2.08$$

A1

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A1

4

[4]

**Q12.**

$$\int_1^3 \frac{1}{\sqrt{1+x^3}} dx$$

| <i>x</i> | <i>y</i> |
|----------|----------|
| 1        | 0.707(1) |
| 1.5      | 0.478(1) |
| 2        | 0.333(3) |
| 2.5      | 0.245(3) |
| 3        | 0.189(0) |

**SC B1** for all correct expressions but wrongly evaluated

3 correct

B1

*all correct*

B1

$$A = \frac{1}{3} \times 0.5 \left[ y(1) + y(3) + 4(y(1.5) + y(2.5)) + 2(y(2)) \right]$$

*use of Simpson's rule*

M1

$$= 0.743$$

A1

4

[4]



EXAM PAPERS PRACTICE

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## Examiner reports

### Q1.

In part (a), Simpson's rule generally was used very well. Very often the students stated the  $x$  values correctly; the  $y$  values had to be correct fractions (the most common choice), recurring decimals or decimals rounded to four significant figures. Some students did not consider  $x = 0$  or used  $x = 5$  which invalidated the formula. A small number of students transposed the multipliers 4 and 2 in the formula. Students should have used large brackets round the expression to help avoid the common error of only multiplying the first term by  $\frac{1}{3}$ .

In part (b), most students realised that the integral gave rise to a  $\ln$  function and obtained  $\frac{1}{2}\ln(x^2 + 2)$ . The common errors were  $2\ln(x^2 + 2)$  or  $\ln(x^2 + 2)$ . The use of limits and the combination of logarithms were usually done well; so many students who did not earn full marks were able to gain 3 marks. Some students left their final answer as  $\frac{1}{2}\ln 9$ . Those students who used the substitution  $u = x^2 + 2$  were usually successful with this question but some students failed to change the limits.

### Q2.

- (a) Although there were many fully correct solutions, some students lost marks by failing to show that they were working to the required accuracy. It was necessary to use values of  $y$  to at least four significant figures as the answer was required to three significant figures. Students should be encouraged to show their answer to greater degree of accuracy than required and then proceed to give the rounded answer. A few students used only four strips instead of five and this gained no credit.
- (b) It was evident that many students did not understand the concept of the mid-ordinate rule even though they were able to apply the formula to answer part (a). Even though the question stated "With the aid of a diagram" only a verbal response was often given. Many diagrams that were drawn illustrated the trapezium rule.

### Q3.

Simpson's Rule was generally well done; a few omitted  $x = 0$ , thus invalidating the formula; a few reversed the 4 and 2 multipliers; and a few made calculator errors, with  $(1 + 64) = 64$  and  $2(4 + 16) = 20$  being the common ones.

### Q4.

This was well answered by the majority of students. Many fully correct responses were seen. The major error was from students who worked in degrees, but they usually worked to the correct degree of accuracy and hence earned the initial B mark for their mid-ordinates and the special case mark for their final answer.

Students who worked to two decimal places (rounded or truncated) were liable to lose the last accuracy mark, but often lost more through not showing enough working.

**Q5.**

In this question those candidates who used radians usually went on to earn full marks. Many candidates, however, used degrees without showing unsimplified correct expressions for  $y$ , and hence only obtained the B mark for 4 correct  $x$ -values.

**Q6.**

Part (a) was very well answered by the majority of candidates. Some candidates lost a mark through not giving the answer to the required degree of accuracy. Candidates losing the final mark had sometimes lost the previous mark by showing their working to insufficient accuracy (usually 3sf). Few attempts at anything other than the mid-ordinate rule were seen.

In part (a)(i), most candidates finished with the required expression though not all had derived it through valid means.

Most candidates obtained some marks on part (a)(ii), and many fully correct responses were seen. The final A mark was often lost because  $dy$  had been omitted. Other errors

occurred during the integration, with  $e^y - 5x$  being very common and  $\frac{e^y}{y} - 5y$  also being popular.

**Q7.**

(a) This was well answered by the majority of candidates with many obtaining full marks. Some candidates lost the final accuracy mark through premature rounding. Most candidates obtained the first B mark and the second B mark although substitution into  $1/(1 + x^3)$  was seen quite often. The main error within the brackets for Simpson's rule was to have the 2 and the 4 reversed, although this was not common.

(b) Most candidates who realised the numerator was related to the derivative of the denominator and hence required a  $\ln$  function were successful in obtaining all 4 marks. Three marks were obtained by candidates who failed to evaluate  $\ln 1$  as 0.

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The major error was the use of an incorrect  $k$  in  $k \ln(1 + x^3)$  with  $k = \frac{1}{2}$  being the most common. These candidates usually obtained both method marks. Many candidates scored zero on this part as they obtained answers by attempts at

integrating  $\int x^2 (1 + x^3)^{-1}$  obtaining expressions involving  $x^3/3$ .

**Q8.**

This was well answered by the majority of candidates giving them a good start to the paper, although many candidates did not apply brackets accurately. Common errors were

to write the answer 2.619 or 2.6195 or to incorrectly evaluate  $\frac{1}{3} \times 2$  as  $\frac{1}{6}$  leading to a final answer of 0.655. The 2 and the 4 inside the bracket were sometimes reversed.

**Q9.**

Many candidates lost marks on this question from careless work. Some candidates scored very well on this question and full marks were not uncommon amongst the more able candidates.

Many candidates obtained full marks in part (a). Where candidates scored 4 out of 5 it was generally because at no stage did they indicate ' $dy$ ' and hence lost the B mark. A

very common error was in  $\int (100 - y^2) dx$ , where the result was often  $(100x - y^3/3)$ , again with candidates very unsure as to whether they were integrating w.r.t.  $x$  or  $y$ . Many special cases were seen and, where attempted, candidates often achieved 2 marks. Several candidates found it difficult to isolate  $x^2$ .

Although many candidates answered part (b) correctly they often lost the final mark for not writing the answer to 3sf. A common error was also to miss out one of the  $x$  values, usually the 0.5. Candidates still seem not to understand that if they require an answer to 3sf then they should be showing either exact values of  $y$  or working to 4 sf.

### Q10.

Most candidates produced the correct derivative in part (a). The main error was the omission of the 3. Substitution of  $x = \ln 2$  was usually correct although there were several cases of  $e^{3\ln 2} = 6$ .

Part (b) was generally very well answered. The majority of candidates attempted the mid-ordinate rule with many fully correct responses seen. Errors occurred with working with 3 s.f. and writing the final answer to an inappropriate degree of accuracy.

Part (c) was done very well up until the last step where a very large number of candidates thought that  $2 - 1/3 = -5/3$  and failed to gain the final mark.

### Q11.

This question was well answered with many candidates gaining full marks. There were quite a few candidates who lost marks by answering to 3 decimal places instead of to 3 significant figures. Some candidates lost marks by using 1, 2, 3, 4, 5 as the values of  $x$ , rather than 1.5 etc, but very few candidates chose alternative methods such as Simpson's rule or the trapezium rule.

### Q12.

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This was reasonably well answered, although many candidates used 3 sf throughout their working and lost the final accuracy mark by giving an answer of 0.742. There were some candidates who used an even number of ordinates.